

1 Executive Architecture Summary Brief

Architecture revision: REV-000003 | **Generation:** DOC-GEN-000002 | **Status:** Machine-generated review draft

Decision boundary: Architecture review material only. Customer issue requires the accompanying approval record. Procurement and installation authority: **No**.

2 Table of Contents

- Executive Summary
 - Domain Direction
 - Overall System Architecture
 - Storage Attachment Topology
 - Risk and Decision Readiness
-

2.1 Architecture-Derived Delivery Summary

This concise summary is derived from the immutable Project Delivery Planning Compiler result. The standalone Project Delivery Plan contains the complete task, dependency, gate, risk and handoff registers.

- Plan maturity: **Indicative**
- Delivery structure: 14 phases, 23 workstreams and 285 tasks
- Governed dependencies: 286
- Risks and engineering handoffs: 0 risks / 71 handoffs
- Indicative completion: Not calculated
- Authority: Architecture-derived planning review only; not a contractual programme, resource-feasibility commitment, procurement authorization or installation authorization.

2.2 Architecture at a Glance

- The selected compute platform is HGX B300.
- The architecture provides 1,152 GPUs.
- The east-west compute fabric uses InfiniBand.
- Plan maturity: **Indicative**

2.3 Network and Connectivity

- Scale-up interconnect uses NVIDIA NVLink / NVSwitch for GPU-to-GPU communication inside each compute domain.
- East-west fabric uses InfiniBand for scale-out workload traffic.
- Storage fabric uses Spectrum-X Ethernet for storage client and system traffic.
- Frontend and north-south fabric uses Ethernet for service and tenant ingress and egress.

- External connectivity: Unresolved — engineering input required
- Out-of-band management fabric uses Ethernet for management, administration and recovery traffic.

2.4 Storage

- WEKA Core AI is represented as a governed architecture review candidate.
- Calculated usable capacity is 4640.91 TB.
- The planning configuration contains 48 WEKA backend nodes.
- WEKA storage throughput: Pending storage engineering / vendor sizing
- Requested usable storage capacity is 4.0 PiB.
- Storage-system power: Not yet calculable — storage configuration required
- Storage fabric demand: Not yet calculable — storage performance and topology inputs required

2.5 Security

- Security architecture guidance includes 2 governed security views.
- Security findings and implementation actions require human review and do not constitute certification or control-effectiveness evidence.

2.6 Architecture Alternative Considered

2 material architecture differences identified.

		Lintasarta 1,024 GPU Physical governed alternative	Main Architecture
#	Parameter		
1	Requested usable storage capacity	4.5 PiB	4 PiB
2	Storage model	High-performance storage with context-memory storage	High-performance storage

2.7 Essential Diagram Commentary

2.7.1 Overall architecture

- The selected compute platform is HGX B300.
- The architecture provides 1,152 GPUs.
- The east-west compute fabric uses InfiniBand.

2.7.2 Principal network

- Scale-up interconnect uses NVIDIA NVLink / NVSwitch for GPU-to-GPU communication inside each compute domain.
- East-west fabric uses InfiniBand for scale-out workload traffic.
- Storage fabric uses Spectrum-X Ethernet for storage client and system traffic.

- Frontend and north-south fabric uses Ethernet for service and tenant ingress and egress.
- External connectivity: Unresolved — engineering input required
- Out-of-band management fabric uses Ethernet for management, administration and recovery traffic.

2.7.3 Storage architecture

- WEKA Core AI is represented as a governed architecture review candidate.
- Calculated usable capacity is 4640.91 TB.
- The planning configuration contains 48 WEKA backend nodes.
- WEKA storage throughput: Pending storage engineering / vendor sizing
- Requested usable storage capacity is 4.0 PiB.
- Storage-system power: Not yet calculable — storage configuration required
- Storage fabric demand: Not yet calculable — storage performance and topology inputs required

2.7.4 Security architecture

- Security architecture guidance includes 2 governed security views.
- Security findings and implementation actions require human review and do not constitute certification or control-effectiveness evidence.

2.8 Review Identity

- **Recorded by:** BD Independent SME Reviewer
- **Role:** Architecture approver
- **Recorded at:** 2026-09-01T10:53:43Z



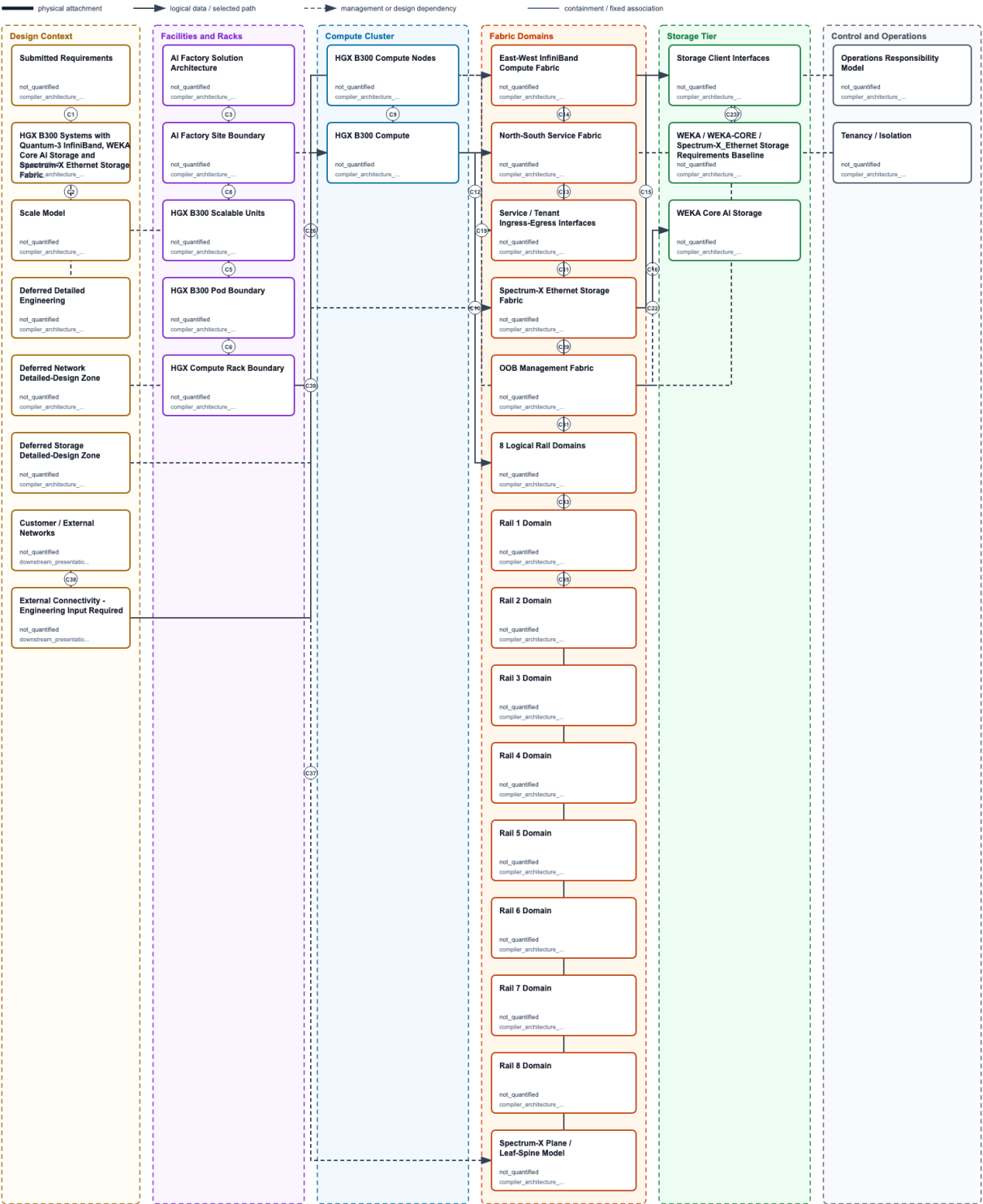
2.10 Figure 2: Conceptual Physical Topology

Neon AI Factory Architect | Solution Architecture

Conceptual Physical Topology

Candidate visual projection. Quantities, media and authority are copied from compiler artefacts or shown unresolved.

source_run_id=RUN-DETERMINISTIC-659a35e | diagram_set_id=DS-ET-RUN-DETERMINISTIC-659a35e-CONCEPTUAL-TOPOLOGY-V1
diagram_id=DS-ET-RUN-DETERMINISTIC-659a35e-CONCEPTUAL-TOPOLOGY-V1-VIEW-CONCEPTUAL-PHYSICAL-TOPOLOGY-R0 | maturity=Production_Grade_Solution_Architecture | generated_at=2026-09-01T10:52:48Z

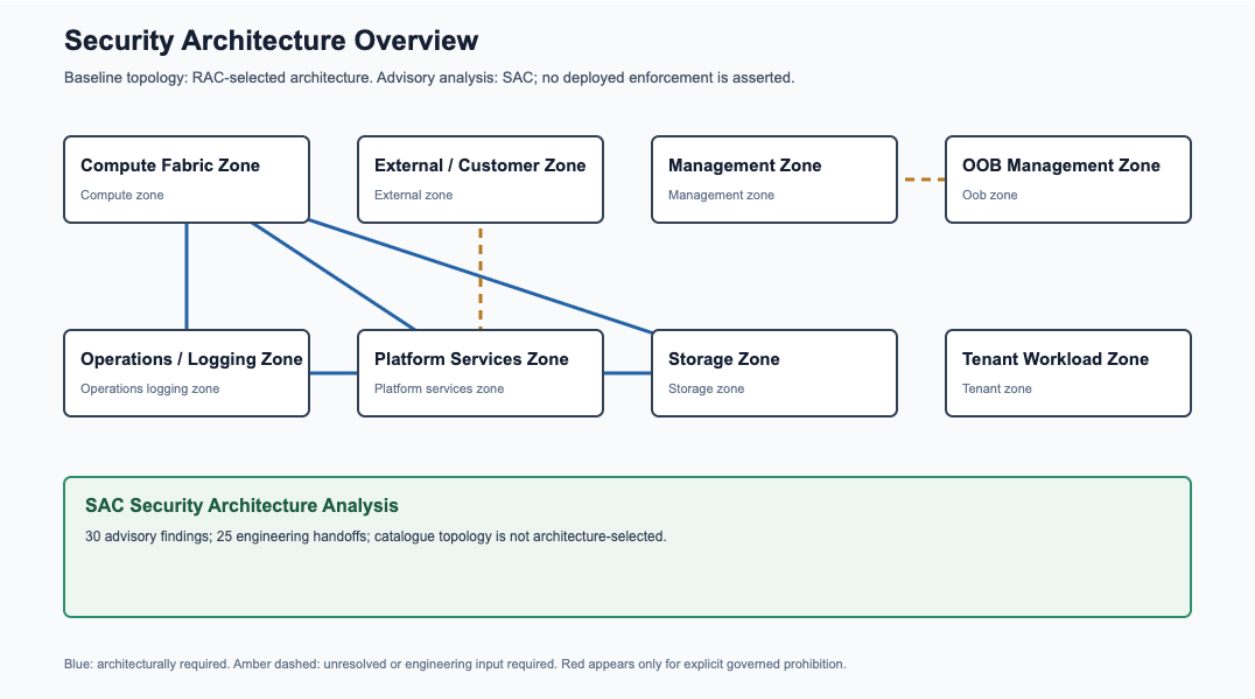


Connection schedule

- C1 design_dependency: compiled into
- C2 design_dependency: selects scale point
- C3 containment: contains deployed system
- C4 containment: contains
- C5 containment: groups
- C6 containment: contains racks
- C7 containment: houses
- C8 design_dependency: defines compute estate
- C9 logical_data_flow: forms GPU compute layer
- C10 logical_data_flow: logical rail attachment abstraction
- C11 logical_data_flow: east-west fabric rail domains
- C12 logical_data_flow: workload service interfaces
- C13 logical_data_flow: workload service and tenant ingress/egress
- C14 logical_data_flow: storage client interfaces
- C15 logical_data_flow: storage client path
- C16 logical_data_flow: core AI storage platform attachment
- C17 logical_data_flow: storage requirements inform client path
- C18 management_control: control / admin
- C19 management_control: compute management plane
- C20 management_control: fabric management plane
- C21 management_control: storage fabric management plane
- C22 management_control: storage system management plane
- C23 design_dependency: fabric isolation policy
- C24 design_dependency: tenant ingress/egress policy
- C25 design_dependency: outside current compiler depth
- C26 design_dependency: detailed network design required
- C27 design_dependency: storage detailed design required

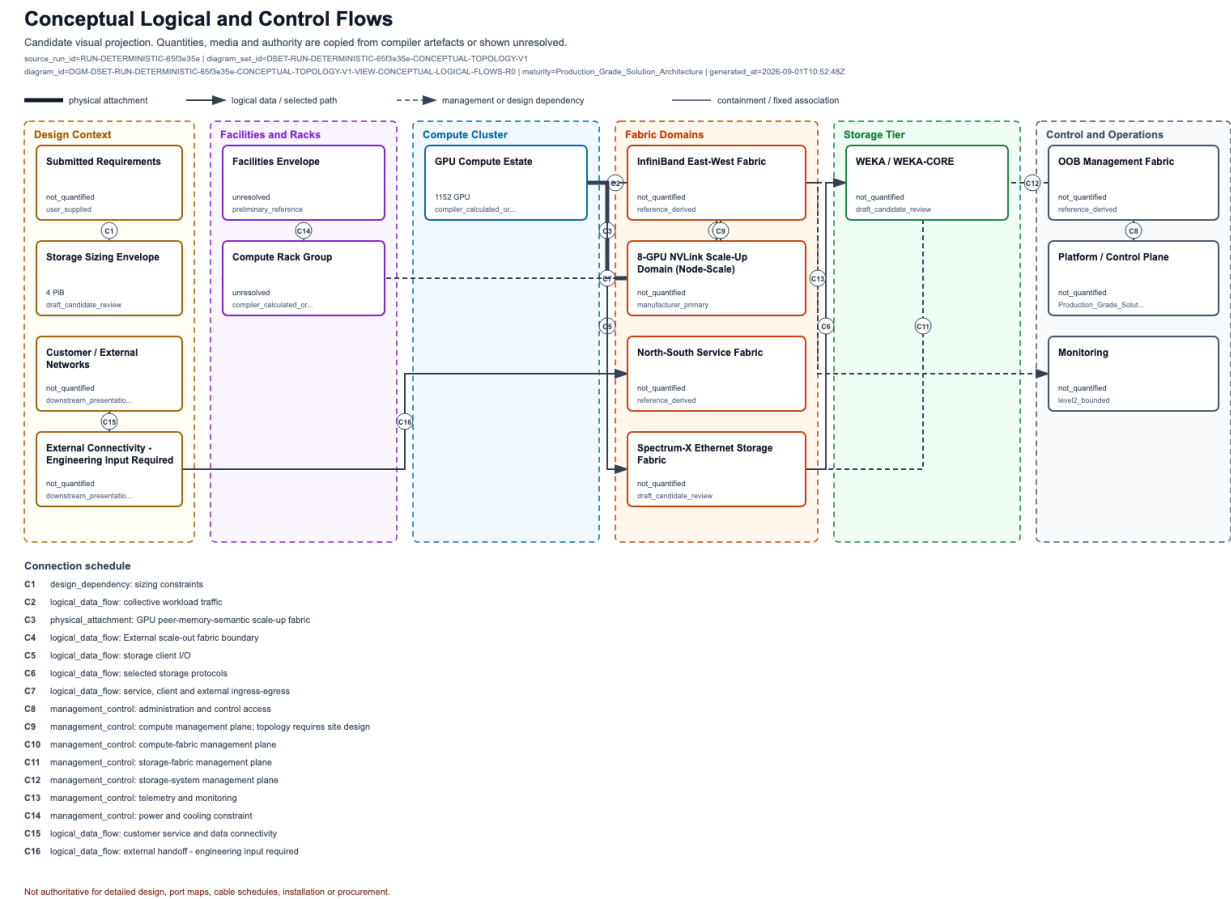
Conceptual Physical Topology

2.11 Figure 3: Security Architecture Overview



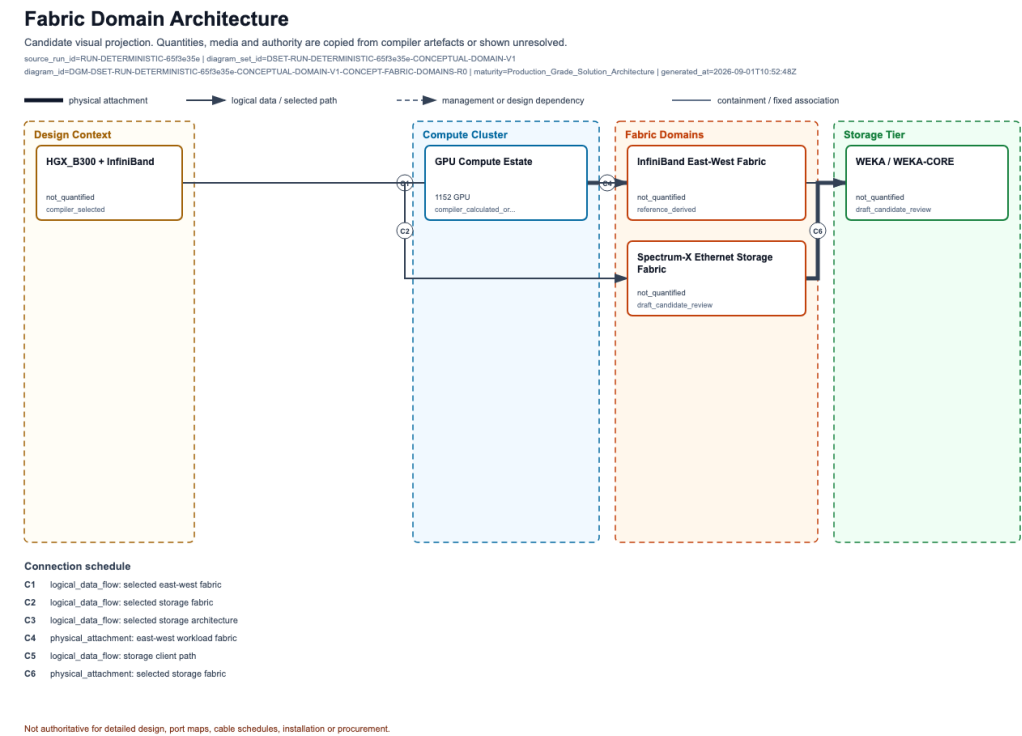
Security Architecture Overview

2.12 Figure 4: Conceptual Logical and Control Flows



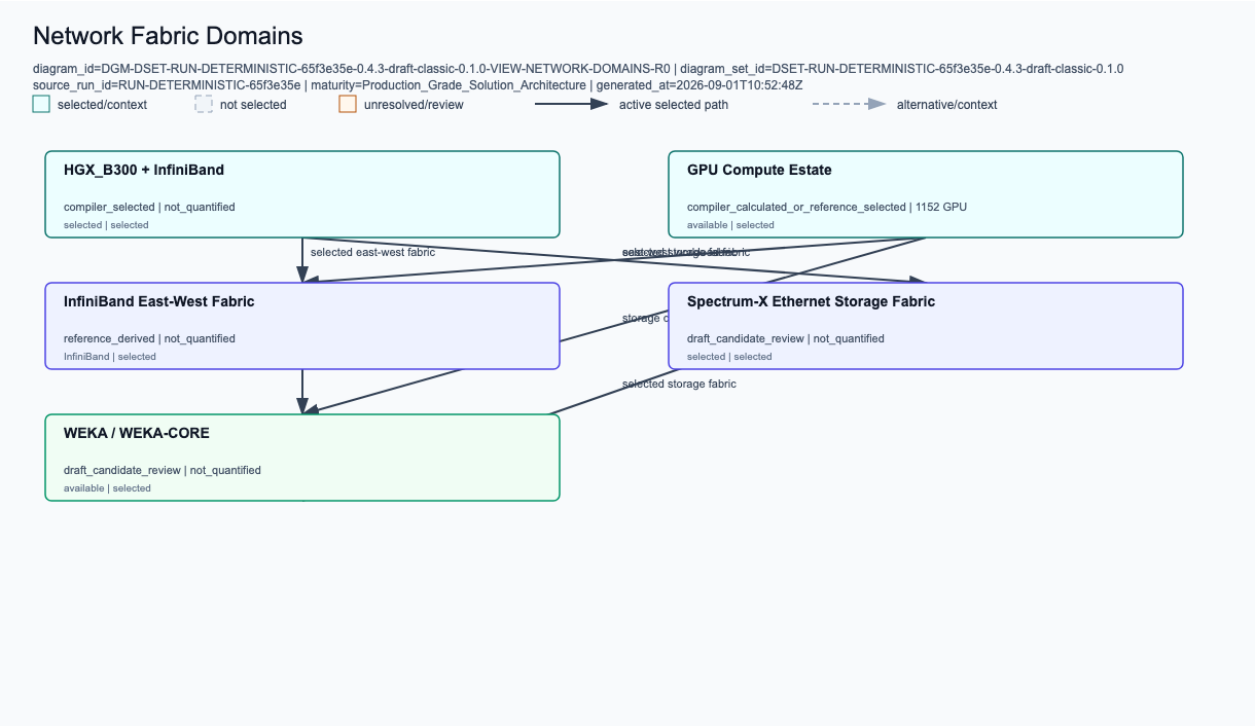
Conceptual Logical and Control Flows

2.13 Figure 5: Fabric Domain Architecture



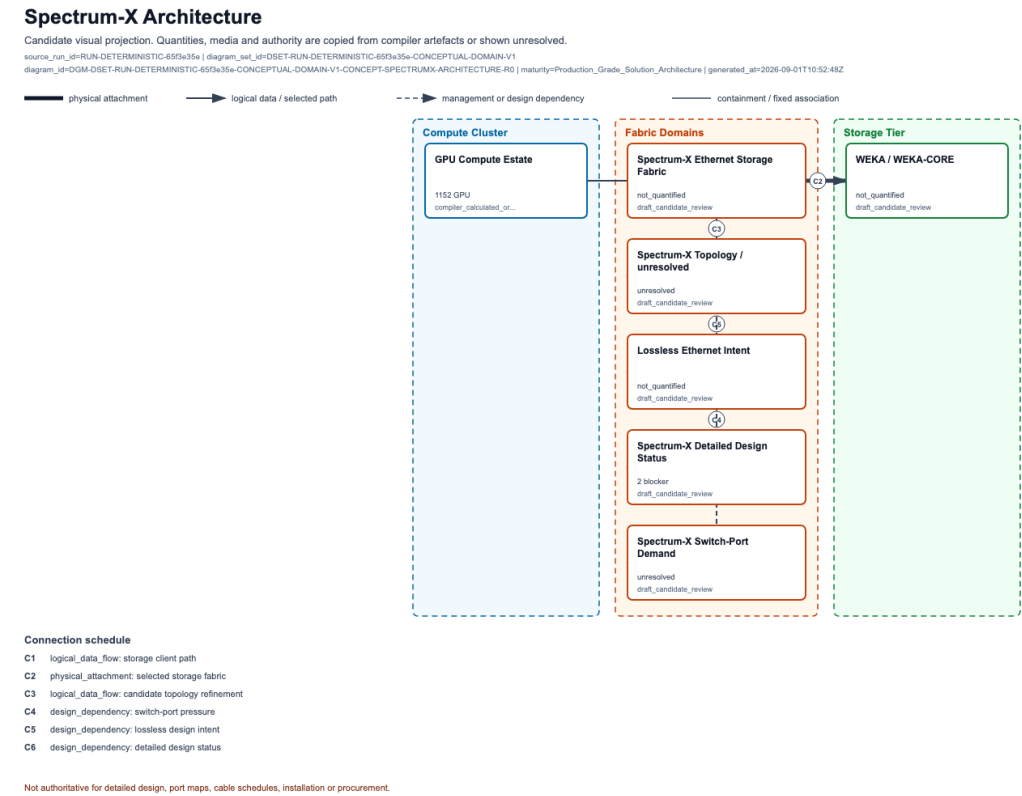
Fabric Domain Architecture

2.14 Figure 6: Network Fabric Domains



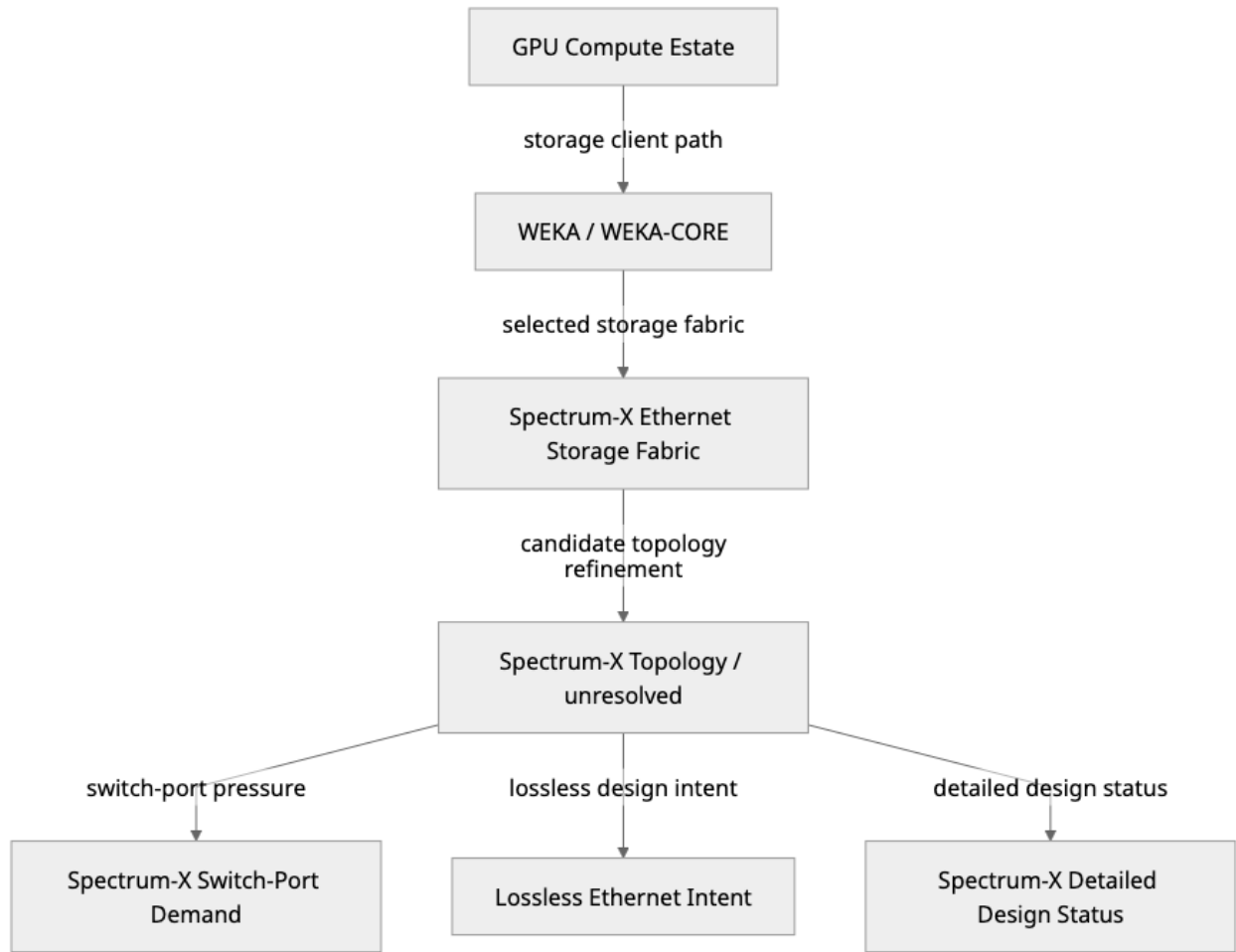
Network Fabric Domains

2.15 Figure 7: Spectrum-X Architecture



Spectrum-X Architecture

2.16 Figure 8: Spectrum-X Logical Topology



Spectrum-X Logical Topology

2.17 Vendor Options

The Vendor Options Compiler presents the governed implementation-option view for this exact architecture revision.

2.17.1 Executive Vendor Options Summary

Architecture area	Current architecture	Compatible implementations	Alternative architectures	Conclusion
Compute	NVIDIA GB300 NVL72	None established	None represented	The governed catalogue currently represents the selected architecture basis only.
Network	Architecture requirement represented	NVIDIA Spectrum-4 SN5600	Arista 7060X6-64PE, Cisco N9364E-SG2-O, NVIDIA	Governed candidates are represented; further

Architecture area	Current architecture	Compatible implementations	Alternative architectures	Conclusion
			Quantum-X800 / Q3400 family	technical evidence is required for some options.
Optical Connectivity	Architecture requirement represented	None established	None represented	Governed candidates are represented; further technical evidence is required for some options.
Storage	VAST Data Platform	None established	DDN AI/HPC storage platform, WEKA Data Platform / WEKApod family	The current basis is retained and alternative architecture options are visible for review.

- Detailed vendor-option requirements: 15; unresolved evidence or sizing items: 15. The full architecture document retains the detailed comparison.

Vendor Options identifies governed compatible implementations and architecture alternatives based on the knowledge available for the selected architecture revision. Some alternatives may require further technical validation or independent branch sizing. Inclusion does not constitute procurement approval, commercial recommendation, availability confirmation, or vendor commitment.

2.18 Phased Physical Realization

This section is calculated from the accepted architecture intent and a governed allocation profile. It provides engineering review evidence, not procurement or installation authority.

The RAC-selected reference fabric supports **144 systems / 1,152 GPUs**. The phased deployment plans an active target population of **128 systems / 1,024 GPUs**, leaving 16 supported node positions unpopulated. Installed fabric capacity is not presented as active GPU population.

Phase	Compute systems	GPUs	WEKA backends	Quantum-X800 leaf/spine	Spectrum-X leaf/spine	WEKA usable capacity
Foundation	8	64	12	8 / 4	2 / 0	1086.17 TB
Expansion	16	128	24	8 / 4	2 / 4	2172.34 TB

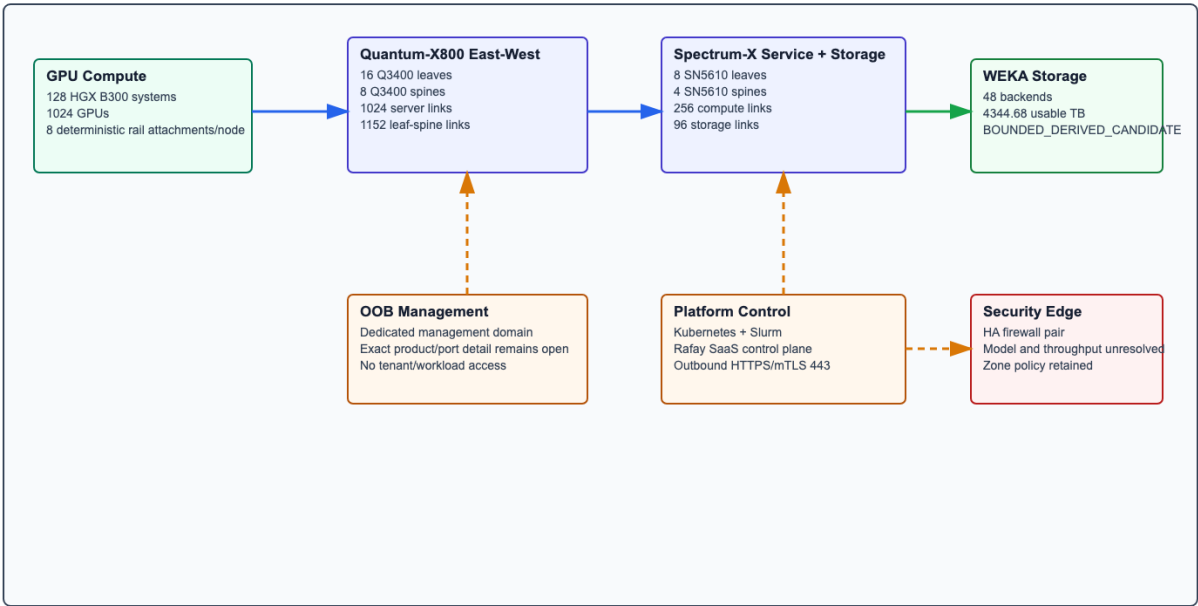
Phase	Compute systems	GPUs	WEKA backends	Quantum-X800 leaf/spine	Spectrum-X leaf/spine	WEKA usable capacity
Target	128	1024	48	16 / 8	8 / 4	4344.68 TB

2.18.1 Overall Phased AI Factory Physical Realization

2784 physical ledger links reconcile to this grouped view. SVG digest:
sha256:e511ed58c07f0ce5fa952f2c6d4f5f65218a23d1e69041c6051cbac4c91586b2.

Overall Phased AI Factory Physical Realization

Target | 1024 GPUs | exact ledger references retained



Blue: workload | Green: storage | Amber dashed: management/control | Detailed ports and every link remain in the machine ledger.

Overall Phased AI Factory Physical Realization

The engineering evidence set contains **143 deterministic vector drill-down views** with a canonical link/device/port index (sha256:828ce49182ce3b6975885b63699cdff9910d4e34ebd189908544e12b2c878f88). These supporting views retain exact physical traceability without rasterizing every link sheet into this document.

Stable device identities and topology-transition records are retained across phases. WEKA Phase 1 preserves the production-observed consultant-confirmed evidence state; Phase 2 and Phase 3 remain bounded derived candidates pending vendor validation.

2.19 Architecture Decision and Approval Record

This section reproduces the human-entered governance records bound to the exact immutable architecture baseline. It does not grant procurement, installation or customer-issue authority.

Record	Identity
Architecture revision	REV-000003

Record	Identity
Baseline	BASELINE-000001
Architecture decision	ADR-000001
Authorised review	REVIEW-000001

2.19.1 Architect Decision

- **Selected alternative:** ALT-MAIN
- **Proposed engineering use:** High-level Solution Architecture
- **Rationale:** The preferred design is suitable for bounded solution architecture handoff based on deterministic compiler evidence.
- **Recorded by:** BD Independent SME Reviewer

Trade-offs

- Detailed network design remains required
- Vendor confirmation remains required
- Procurement and installation authority are excluded

2.19.2 Authorised Review

- **Decision:** Approved with conditions
- **Rationale:** Accepted as a bounded Solution Architecture candidate after review of requirements, options, diagrams, BOM and handoff dependencies.
- **Recorded by:** BD Independent SME Reviewer

Approval conditions

- Detailed design validation required
- Vendor sizing and commercial quotation required
- Facilities validation required

2.19.3 Bounded Baseline

- **Classification:** Solution Architecture baseline
- **Approved use:** High-level Solution Architecture

Limitations

- Not procurement ready
- Not installation ready
- Detailed design validation required

2.20 Risk and Decision Readiness

- The baseline assessment identified 66 advisory architecture findings.
- Coverage status: **Incomplete**; coverage gaps: 1.
- Findings require human review and do not constitute risk acceptance, certification or control-effectiveness confirmation.